

Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Spring 2025
Laboratory Exercise 2
Inverter Schematic and Simulation

Due: 1/31/2025

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1 Introduction

This exercise will teach you how to design a CMOS inverter, convert custom circuits to symbols, and create a proper testbench for pre-layout simulation of basic CMOS logic gates.

Optional: There are also instructions on how to create a Verilog model (see section 2.3). This part is an artifact left over from previous semesters **and is not required for signoffs.**

Check section 4 for the lab report requirement and section 5 for the required items for signoff. **Please submit your lab report (with the signoff sheet attached) to myCourses**

Disclaimer: The instructions for this lab were made with an older version of Cadence Virtuoso

Keep the following in mind as you work through this lab.

- **NEVER** COPY CADENCE LIBRARY FILES FROM THE COMMAND LINE OR FILE MANAGER
- IF ANY OF YOUR CADENCE FILES ARE LOCKED, CLOSE THE TOOL AND EXECUTE THE FOLLOWING COMMAND:

```
/classes/ee620/bin/unlock.cadence.csh
```

- **Hotkeys:** You should learn how to use the hotkeys for better productivity. You can access a list of hotkeys (also known as bindkey) from CIW (the first window that pops up after you launch virtuoso). In CIW, select the menu option:

Options → Bindkey

Select a category (e.g. **schematic**) and a list of current hotkeys (bindkeys) will be shown for it.

REMINDERS:

- Always start the Cadence tools using the **virtuoso &** command from your cmpe630 directory and NOT from your home directory.
- Always properly close the Cadence tools before you log off, otherwise you WILL lose your work and/or potentially lock your cells and not be able to access them!

2 Procedure

2.1 Create the Schematic

Reminder: 'fml' is an example for first, middle, last name. Please use your initials!

In your previous exercise you learned how to make a schematic. This time you will do it on your own (Don't worry, it's a simple inverter). To start, create a new cell in your personal cell library. Go to the Library Manager window, select your library **fml_lib** and then select **File → New → Cellview**. This time, you will name your cell "FML_INVX1" and **make sure the view is set to schematic**. Continue following the same instructions from the previous lab exercise.

When sizing the transistors, make sure you change the sizing values! **The NMOS has a length of 45nm and total width of 120nm, PMOS has length of 45nm and total width of 240nm.** Check and save your schematic and check the CIW for log messages that your schematic is clean. You should receive a "schematic check completed with no errors" and your final schematic should look like Fig. 1.

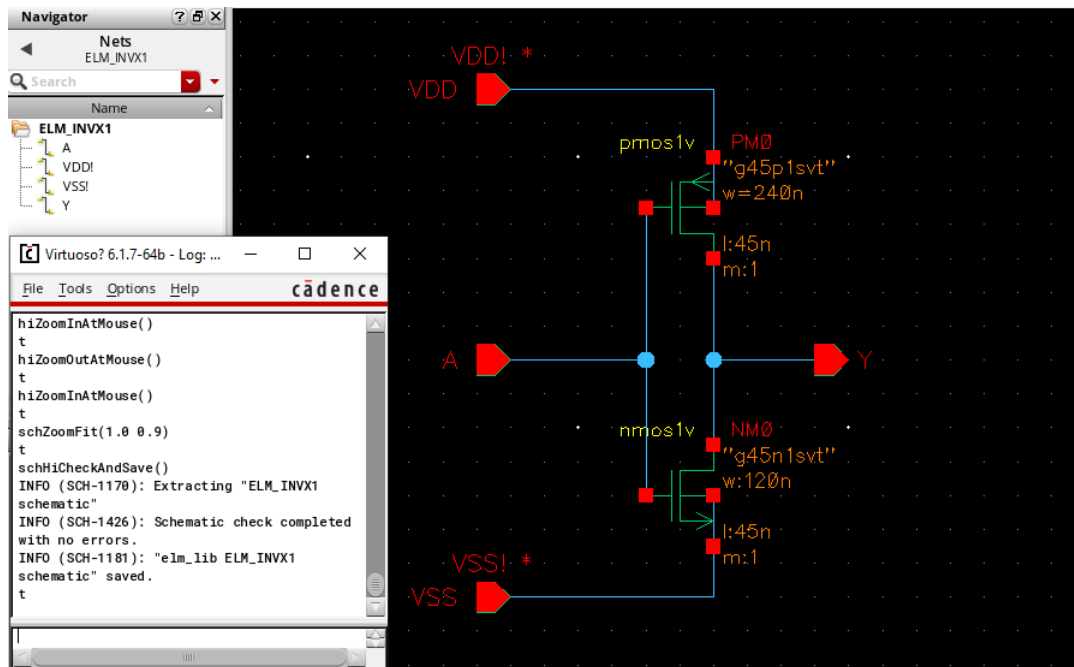


Figure 1: Final Schematic

If you receive warnings or errors, you need to fix these issues before proceeding!

The TAs and lab instructor will be checking your final inverter schematic and clean CIW output message for **Signoff**

2.2 Create a Symbol

Do not close your schematic capture window. Create a symbol view by selecting **Create → Cellview → From Cellview**, and the following (see Fig. 2 should appears).

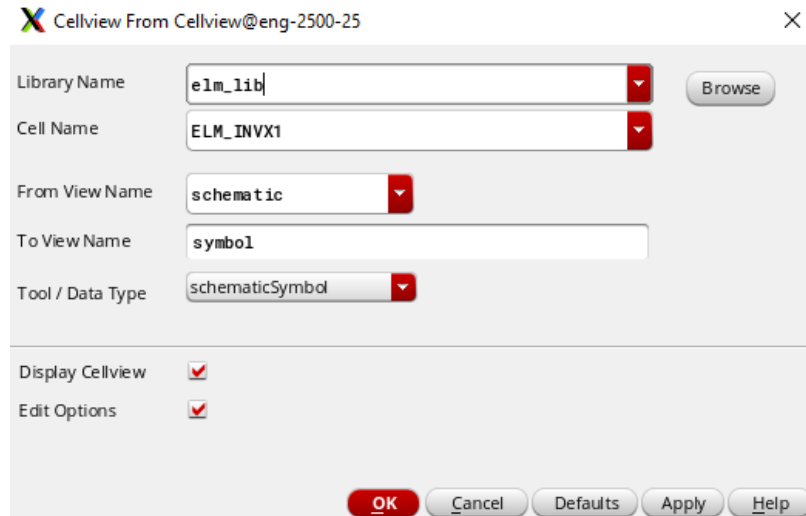


Figure 2: Create Schematic Symbol Cellview from Cellview Form

Click **OK** and an options window will appear, shown in Fig. 3. Here, select the option **All** under **Exclude Inherited Connection Pins** and ensure that VDD and VSS are present within the field. Click **OK** to create the symbol view.

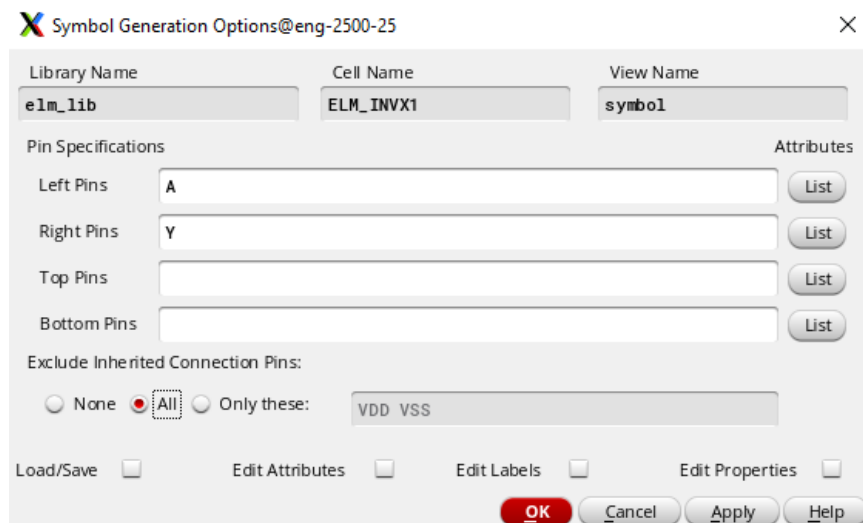


Figure 3: Symbol Generation Options

Now your new symbol will appear as shown in Fig. 4.

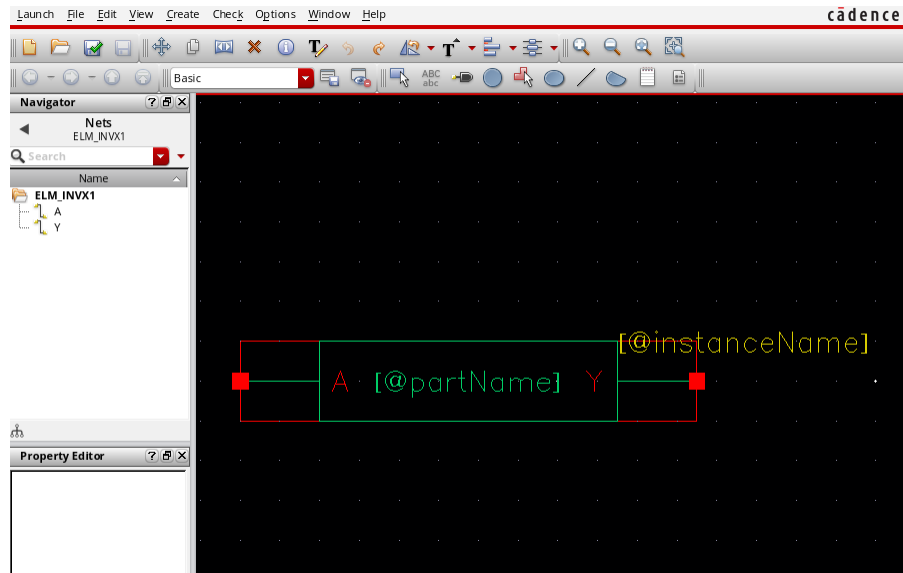


Figure 4: New Symbol Cellview

Edit the symbol to appear more like a gate as shown in Fig.5. ONLY edit the cell outline - the green portion of the symbol - not the symbol parameters. (This will make your final project more understandable and easy to debug).

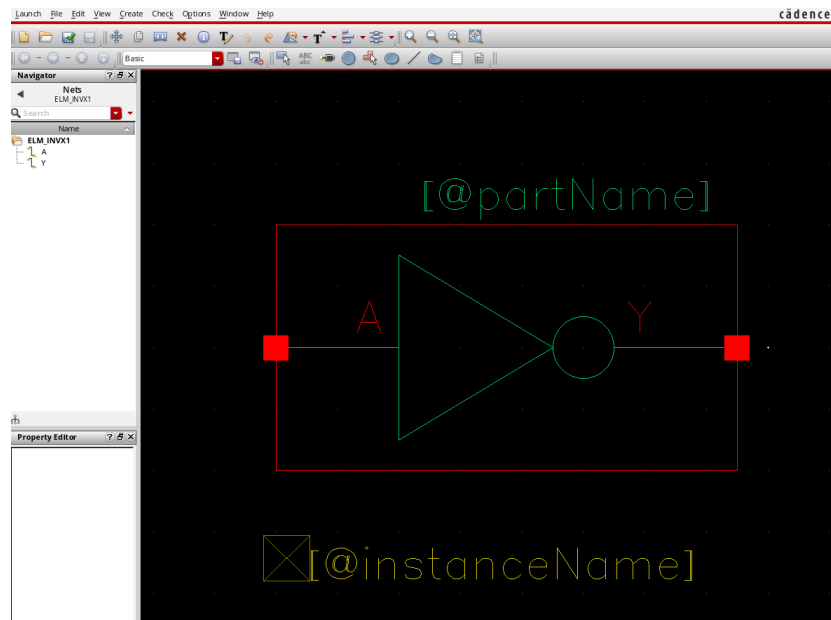


Figure 5: Edit Symbol Cellview

You can now **check and save** and close the symbol. **Note** that you may receive warnings about the missing VDD and VSS pins which you can ignore.

You can use your new symbol by instantiating it in a schematic. Before moving on, close your symbol window.

2.3 Optional - Creating a Verilog Model

This section is optional. Feel free to move on to the next section.

Using your schematic capture window, create a Verilog view by selecting **Create → Cellview → From Cellview**. Make sure that the “From File View” name targets schematic and change the “Tool/Data Type” to Verilog-Editor.



The screenshot shows a dialog box titled "Create Verilog HDL Cellview from Cellview Form". It contains the following fields and controls:

- Library Name:** A text box containing "elm_lib" with a dropdown arrow on the right.
- Cell Name:** A text box containing "ELM_INVX1" with a dropdown arrow on the right.
- From View Name:** A dropdown menu currently showing "schematic". A red arrow points to this dropdown.
- To View Name:** A text box containing "functional".
- Tool / Data Type:** A dropdown menu currently showing "Verilog-Editor". A red arrow points to this dropdown.
- Display Cellview:** A checkbox that is checked.
- Edit Options:** A checkbox that is checked.
- Buttons:** At the bottom, there are buttons for "OK", "Cancel", "Defaults", "Apply", and "Help".

Figure 6: Create Verilog HDL Cellview from Cellview Form

Click OK and a text editor will appear.

NOTE: You may receive a warning message window indicating that your symbol and Verilog module have pin mismatches. This warning can be safely ignored. DO NOT regenerate your symbol to fix the noted pin issues. (The mismatch occurred due to VDD and VSS pins being intentionally not added to your symbol; they do however appear in your Verilog module and schematic. We keep it this way so you can safely use your symbol in other schematics without any VDD and VSS attached.)

Take a look at the pins shown in the module declaration. Your model should have the same pins. Your screen should be identical to the window shown in Fig.7

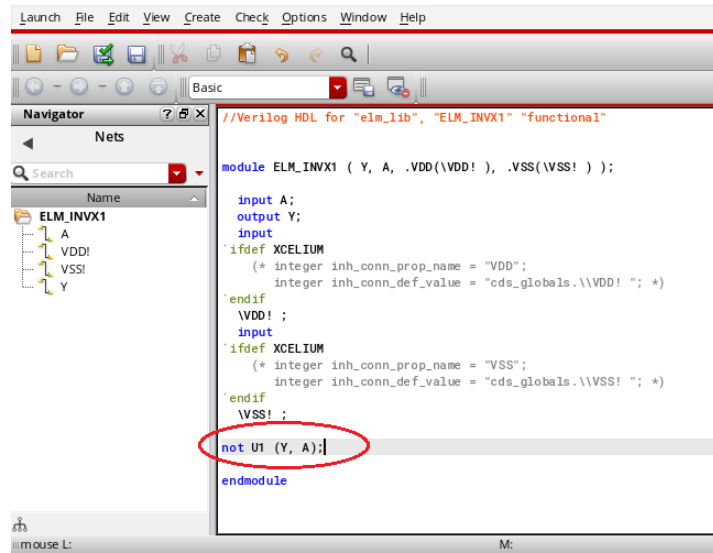


Figure 7: Verilog HDL Cellview

Check and Save the code. You might have same warning mentioned above in “NOTE:”, please follow the same instructions. DO NOT regenerate your symbol to fix the noted pin issues!

You will receive a syntax report in your CIW that should indicate your code is clean as shown below.

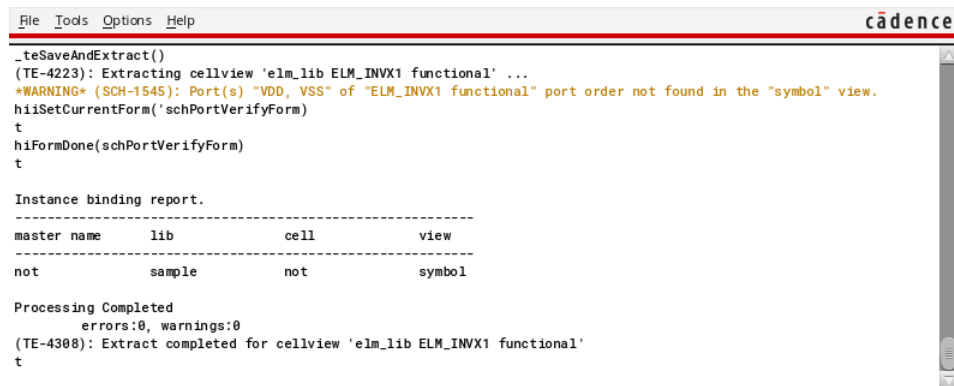


Figure 8: Check and Save Verilog HDL Cellview CIW Log Messages

Close your Verilog model; next we'll simulate your circuit with Spectre, a high performance Spice like simulator.

2.4 Simulate the Circuit (Pre-Layout)

The advanced simulation environment requires a "testbench" for the device under test (DUT or the circuit you wish to simulate). Using the Library Manager, create new schematic cellview and name it FML_INV1_TEST. Design your schematic as shown in Fig.9

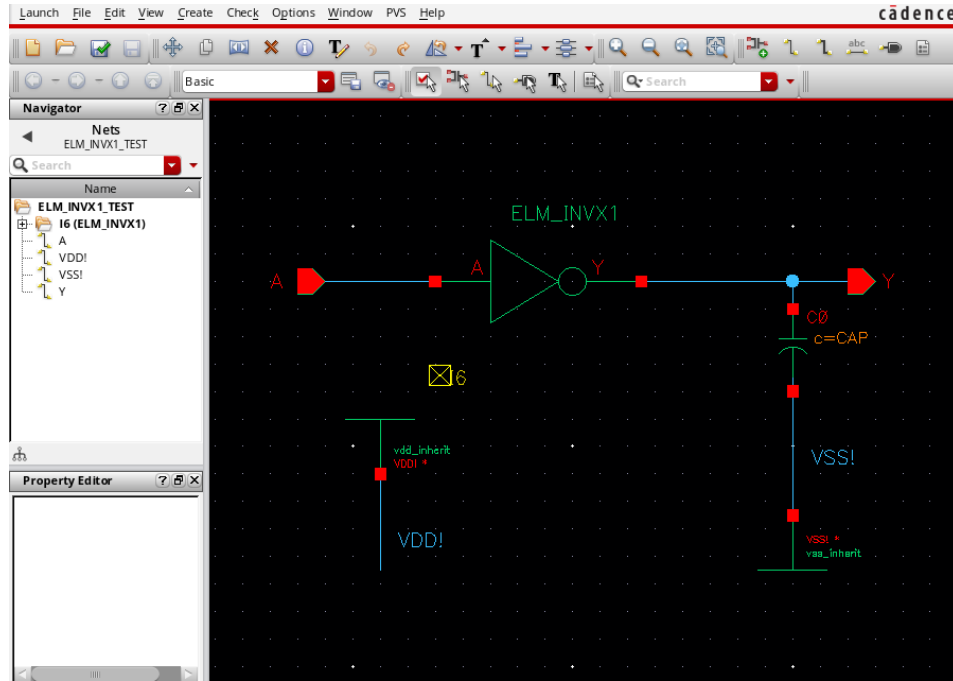


Figure 9: ELM_INVX1_TEST Testbench Schematic

Place one instance of your inverter cell. Next, add a capacitor from the analog library. Click **browse** on the Add Instance and select the analog library **analoglib**, and then the capacitor **cap**. The Add Instance form should update as shown in Fig.10.

Library: analogLib

Cell: cap

View: symbol

Names: C0

☒ Add Wire Stubs at:
☐ all terminals ☒ registered terminals only

Array: Rows: 1, Columns: 1

Buttons: Rotate, Sideways, Upside Down

Model name:

Capacitance: CAP F

Width:

Length:

Multiplier:

Scale factor:

Temp rise from ambient:

Initial condition:

Temperature coefficient 1:

Temperature coefficient 2:

Capacitor Area:

Capacitor Perimeter:

Buttons: Hide, Cancel, Defaults, Help

Figure 10: Add Capacitor instance to Schematic Form

Names: VSS!

☐ Expand bus names

☐ Attach to multiple wires

☐ Create net alias labels

☐ Display bundles vertically

Font: Height: 0.06250, Style: stick, Justification: lowerCenter

Entry Style: fixed offset

Position: Offset: Rotation:

Horizontal Wires: above, 0.04500, R0

Vertical Wires: left, 0.04500, R90

Rotation: Buttons: Rotate, Sideways

Buttons: Hide, Cancel, Defaults, Help

Figure 11: Create Wire Label for VSS Form

We are going to create a **capacitor variable** so we can modify the value for future tests. Modify the Capacitance value to **CAP F**, and hit the tab key. Then click Hide/OK.

Take a look at Fig.11. VDD and VSS need special treatment; add the cells **vdd_inherit** and **vss_inherit** from the analog library. Wire these cells as shown and then add wire labels using **Create → wire → name**. Note that adding the cells is first, wiring is second and adding labels are as the last step. The following shows creating the label VSS!.

Attach the label to the appropriate net. Check and save your schematic. Again, the final product should look like Fig. 9.

2.5 Starting Analog Environment

We'll now start the analog environment, Select **Launch → ADE XL** from your testbench schematic. You'll get the following window:

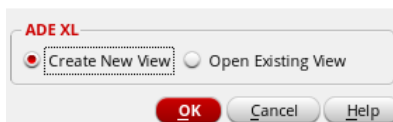


Figure 12: Launch ADE XL

Click OK. Then a window to create the new ADE XL view will appear, shown in Fig.13. Then Click OK.

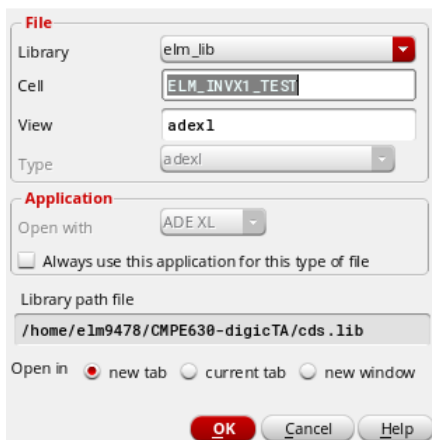


Figure 13: Create new ADE XL View Form

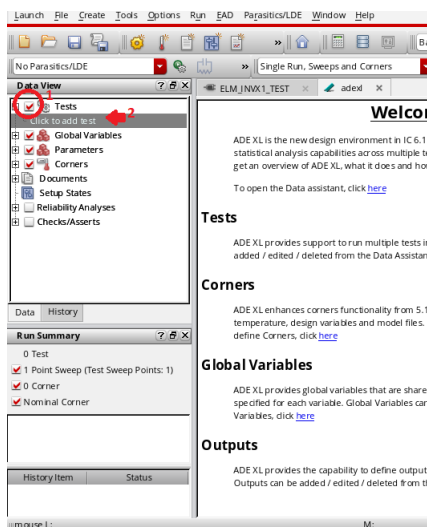


Figure 14: Initial ADE XL Testbench Window

The welcoming page will show up shown in Fig.14. To the left, click on **Tests** and then **"Click to add test"** in the Data View window.

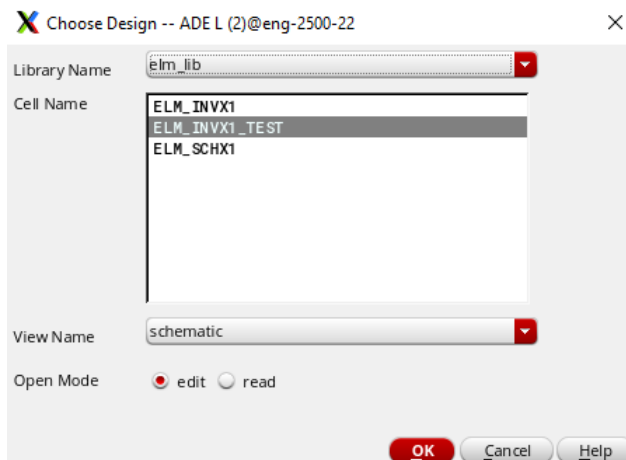


Figure 15: ADE XL Test Editor Window

Once the test editor window opens, you'll be prompted to select your test bench design shown in Fig.15. Now, click OK and you will see the test editor window. (see Fig. 16)

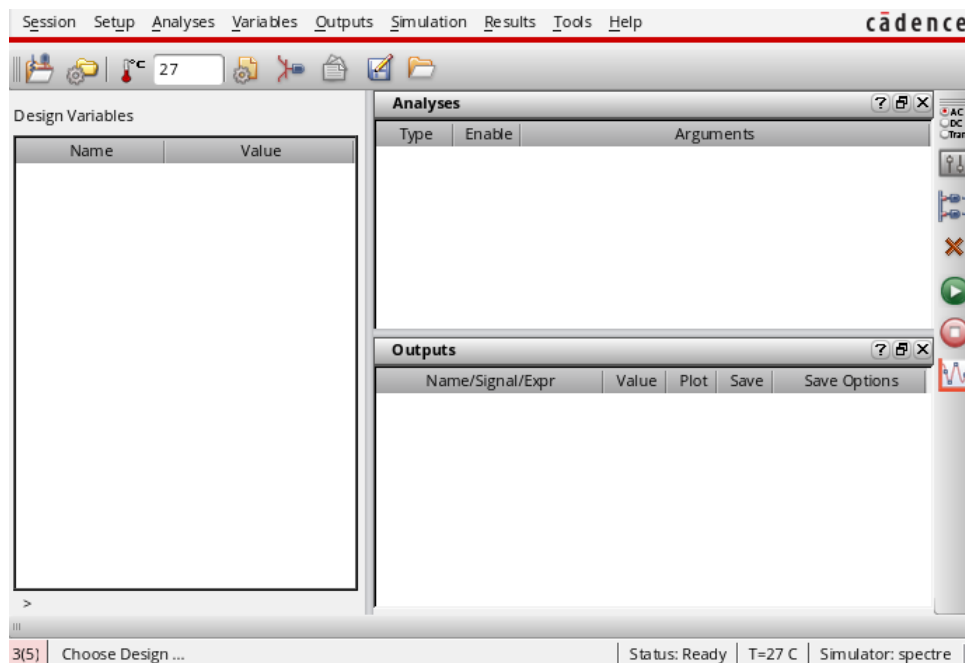


Figure 16: ADE XL Test Editor Window

APS mode is enabled for simulation by default, and we will turn that off. Select **Setup → High-Performance Simulation** and the following window will appear (see Fig 17).

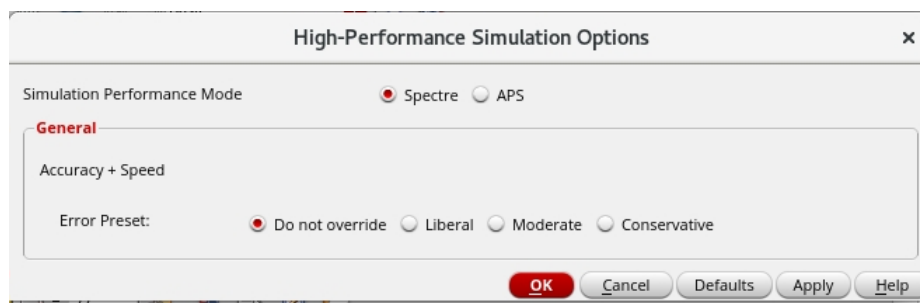


Figure 17: High Performance Simulation Options

Be sure to select **Spectre** for the simulation performance mode, your test results might be invalid otherwise.

We'll then tell the simulator which "section" of the transistor models we are going to use - model files typically have multiple sections representing "corners", such as fast / fast or minimum (ff), typical (tt), and slow / slow or maximum (ss). Select **Setup → Model Libraries** and the following window will appear (see Fig. 18) .

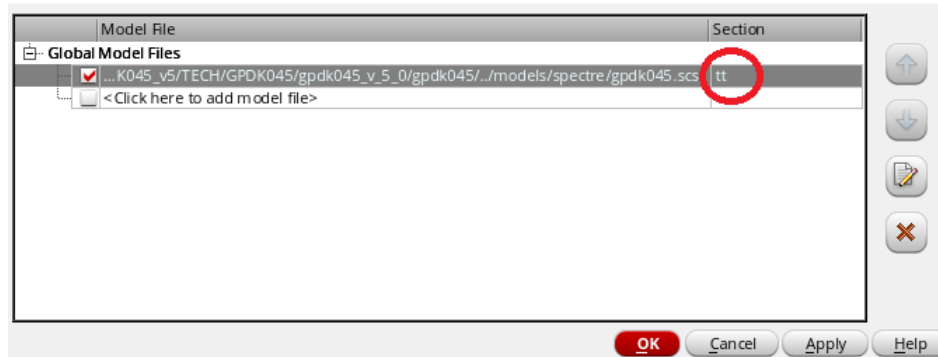


Figure 18: Configure Model Files Form

Be sure to select the model **tt** in the section box. Click OK.

Now, go to **Setup → Stimuli**

Select **Inputs**, enable input A and change its function to **pulse**, make sure **Enabled** is checked. Complete values as shown in Fig.19.

Click **apply**. (If you don't "Apply" your changes wont save and you will encounter errors!) Input A is now enabled!

Now select Global Sources and complete the configuration form as shown in Fig.20.

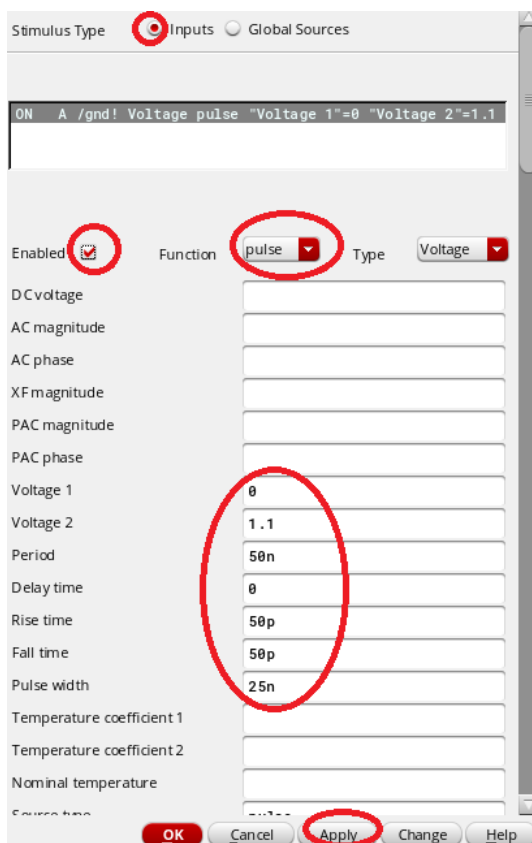


Figure 19: Configure Input Stimulus Form

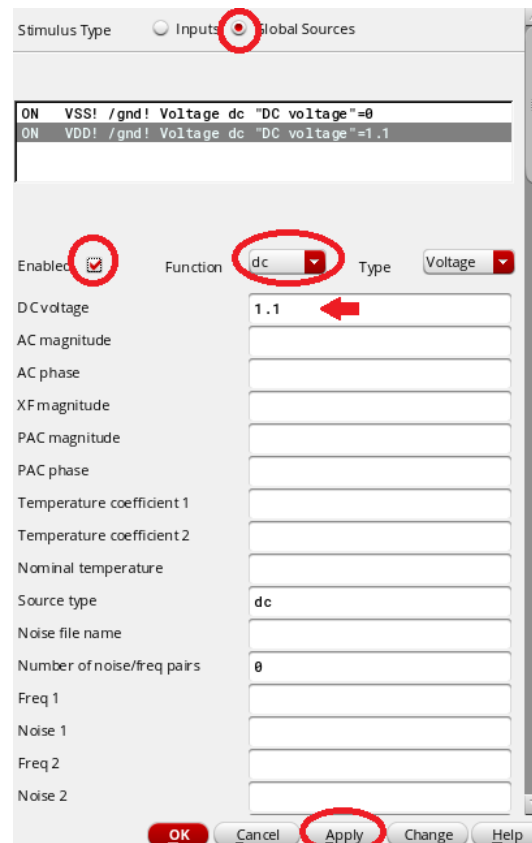


Figure 20: Configure Input Stimulus Form

Click OK to close this selection window.

Now select **Analysis → Choose**, and fill out the next window as shown in Fig. 21

The screenshot shows the 'Configure Analysis Form' dialog box. It is divided into several sections. The top section, 'Analysis', contains a grid of radio buttons for different analysis types: tran (selected), dc, ac, noise, xf, sens, dcmatch, acmatch, stb, pz, lf, sp, envlp, pss, pac, pstb, pnoise, pxf, psp, qpss, qpac, qpnoise, qpxf, qpss, hb, hbac, hbstb, hbnoise, hbsp, and hbxf. Below this is the 'Transient Analysis' section, which includes a 'Stop Time' field set to '200n', 'Accuracy Defaults (errpreset)' with three checkboxes (conservative, moderate, liberal), a 'Transient Noise' checkbox, a 'Dynamic Parameter' checkbox, and an 'Enabled' checkbox with a checkmark. At the bottom of the dialog are five buttons: 'OK', 'Cancel', 'Defaults', 'Apply', and 'Help'.

Figure 21: Configure Analysis Form

Select **tran** (Transient Analysis) and set Stop Time to 200n (200 nanoseconds).

Click **Enabled** and then click OK.

Now we need to select the outputs to be saved and plotted. Select **Outputs → To Be Plotted → Select On Design** (Depending on the context, the menu pick may show as **Outputs → To Be Plotted → Select On Schematic**). In the schematic window, click once on the circuit nodes you want plotted. In this case input A and output Y. You'll notice the nets change colors as shown below (see Fig. 22).

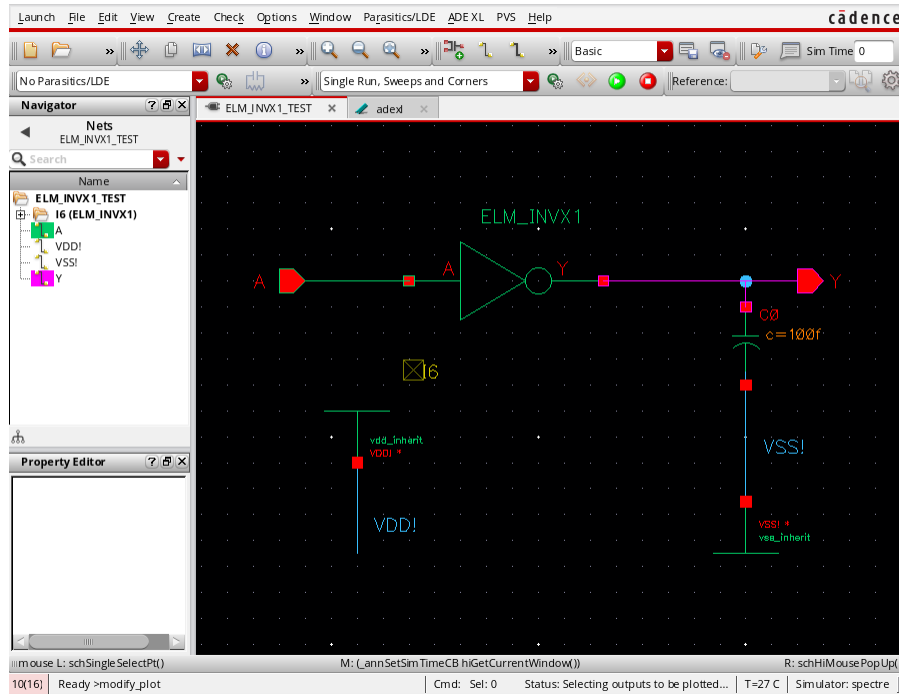


Figure 22: Schematic with Signals Selected for Plotting (ignore the capacitor value, it should be $c=CAP$)

After you make the selections, make sure you press **ESC** to avoid leaving the plot command active in the schematic capture window.

Now, let's add CAP as a design variable (CAP is the capacitance variable we made in the testbench schematic). Underneath the Design Variable window in your test editor, right-click and select **Copy From Cellview** (see Fig. 23). Then enter the 100f as the corresponding value for CAP.

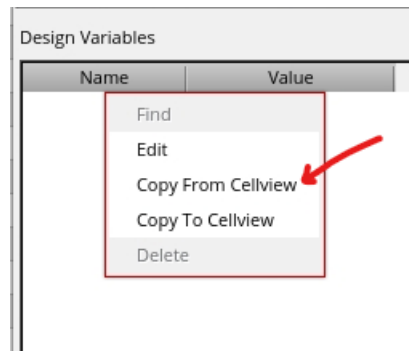


Figure 23: Copy from cell view

Once that's done, your test editor should look like the following (See Fig. 24).

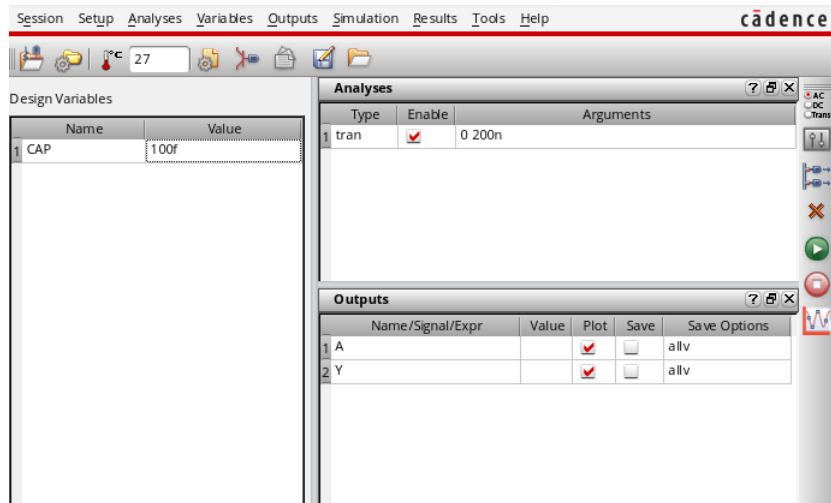


Figure 24: ADE XL Test Editor Window - Configuration Complete

Select **Select Session → Quit** and now you will return to the ADE XL window (Select ‘adexl’ if necessary). Your adexl window should look like this.

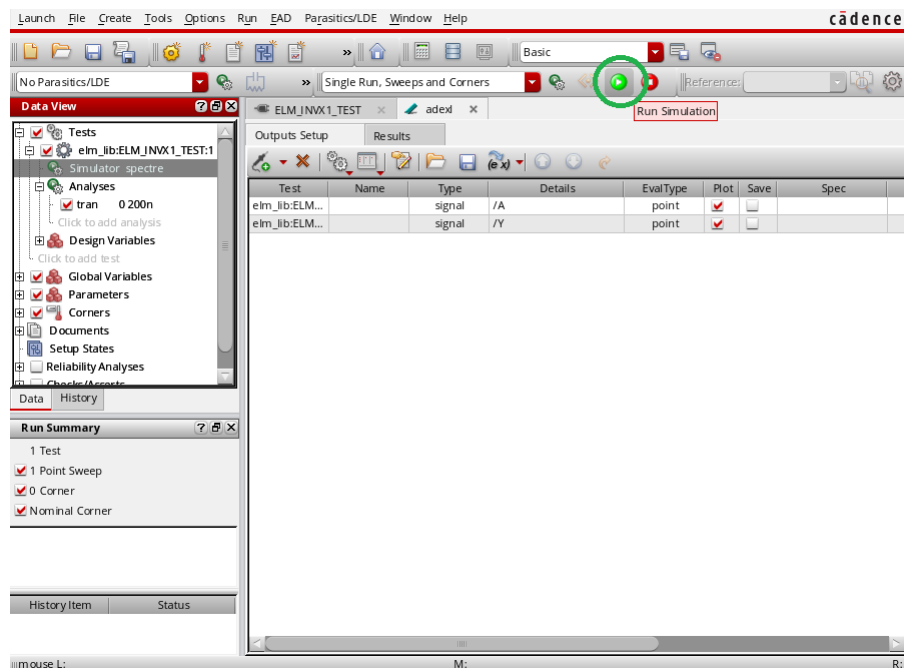


Figure 25: ADE XL Main Window - Configuration Complete

You are ready to run your first transient simulation! Click on the green start icon (see Fig. 25). The tools will generate a netlist of your schematic which is used by the simulator to create the admittance matrix of the circuit. The simulator solves the matrix, then saves and plots the results.

Once your simulation is successful, you'll see new waveform indicators on the results tab as shown below.

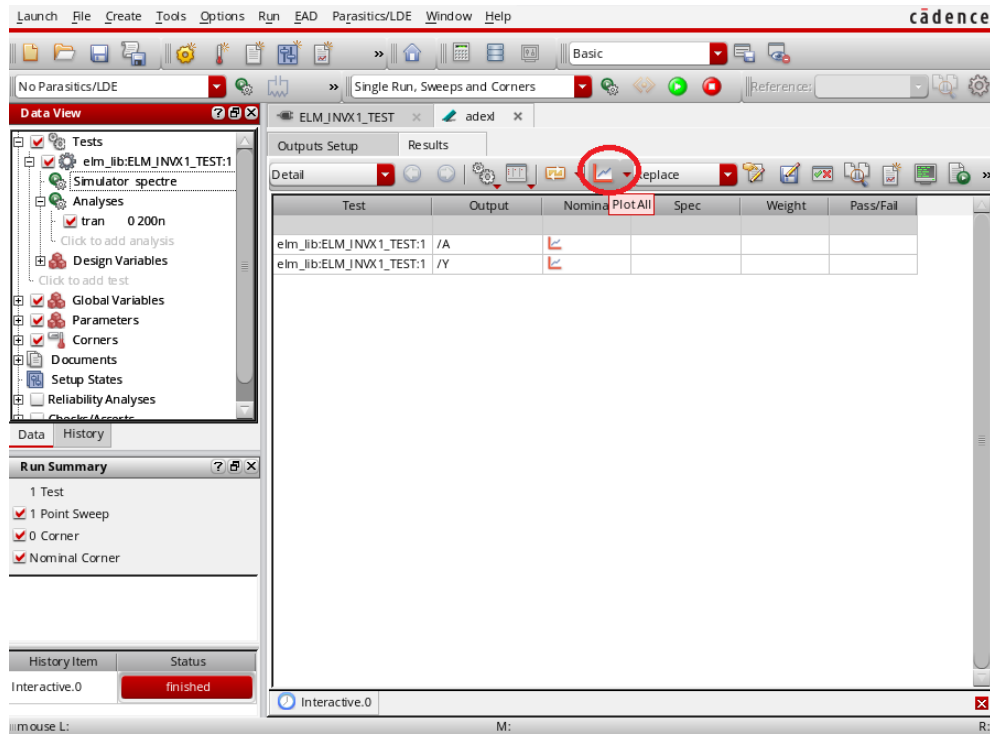


Figure 26: ADE XL Test Editor Window - Simulation Complete

Click on the little graph icon (**Plot all**) as shown in Fig. 26. The waveform window will then be opened as shown in Fig. 27.

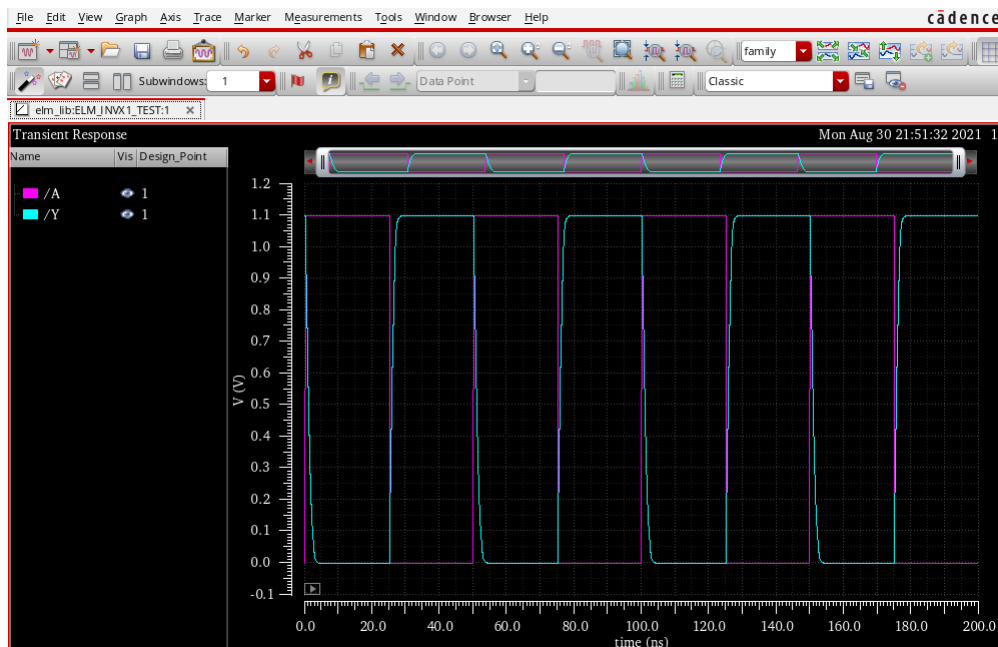


Figure 27: ADE XL Waveform Window - Overlapping Waveforms

The waveforms will initially be overlapped; click on the **“Split All Strips”** icon to separate your waveform into strips as shown below (see Fig. 28):

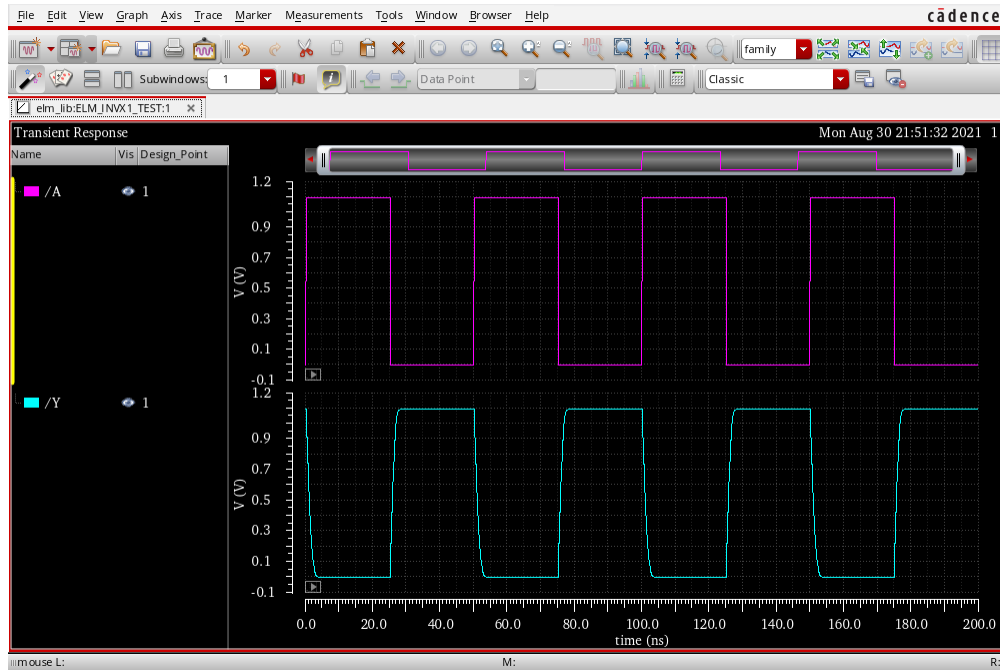


Figure 28: ADE XL Waveform Window - Strip Waveforms

Use the zoom icons to zoom in and out of the displayed area as shown below (see Fig. 29):

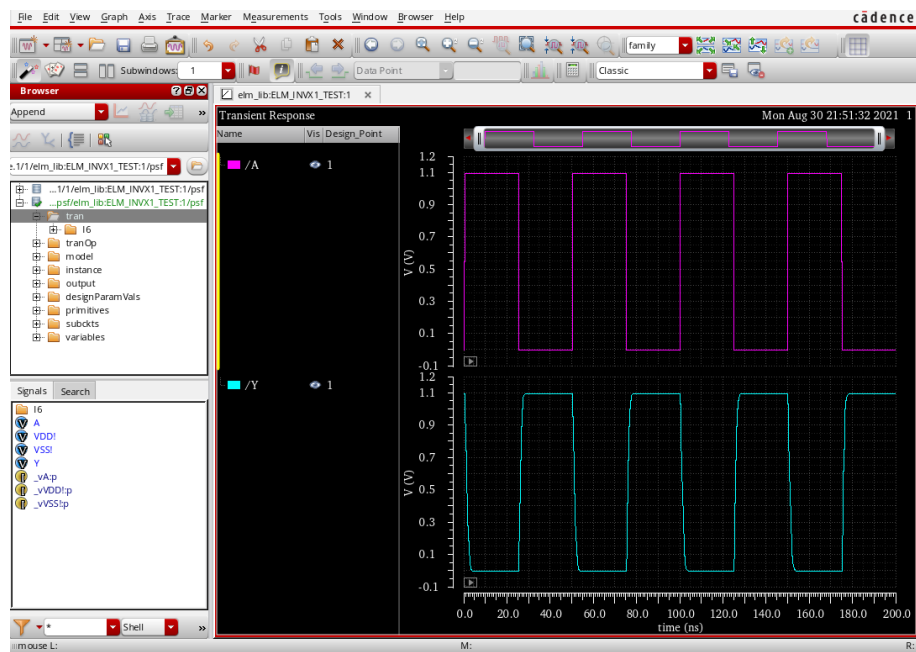


Figure 29: ADE XL Waveform Window - Zoom Into Waveforms

This is the transient simulation signoff. Please also do take a screen capture of some waveform plots for your report (You'll thank yourself later).

2.6 Example For Test Measurements

The ADE XL Waveform viewer provides a powerful suite of tools to perform analysis and calculations using the simulator outputs. From the waveform window, open the design Browser using the menu pick **Browser → Results → Open Results**. Select the PSF database as shown below (see Fig. 30).

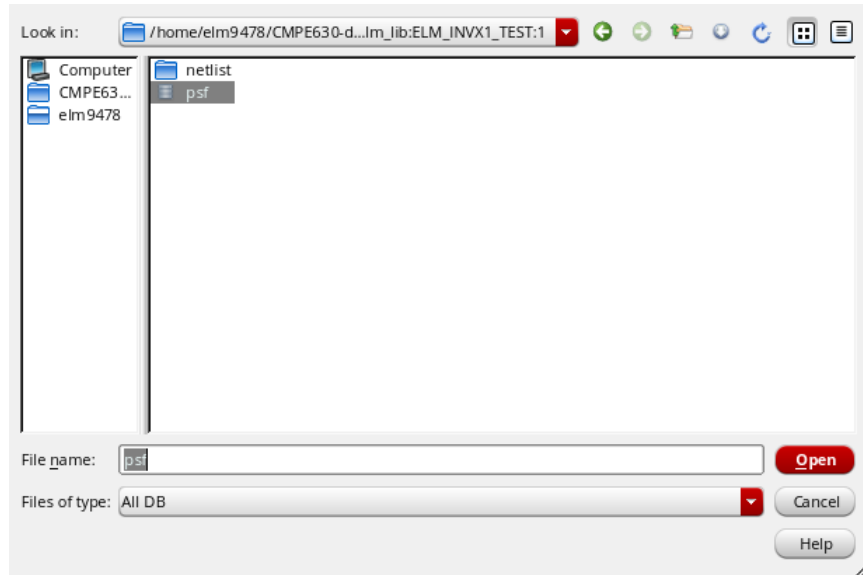


Figure 30: ADE XL Select Waveform Database Form

Click Open.

The Browser panel opens on the left of the waveform window; using the upper browser panel, navigate and select **tran** as shown in Fig. 31. Signals will appear in the lower browser panel, select the output Y, right click, and open the calculator (again, see Fig. 31).

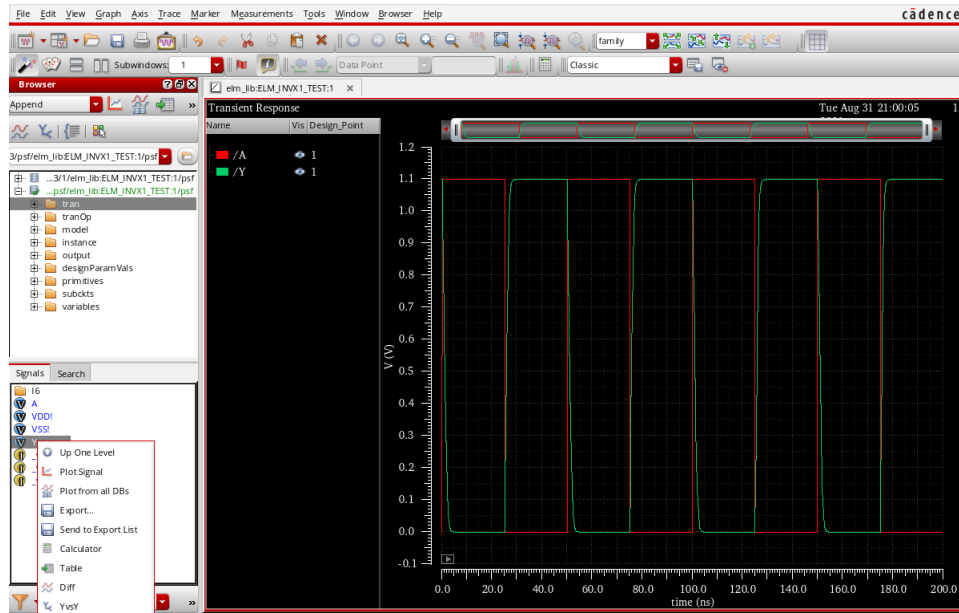


Figure 31: Waveform Window with Hierarchy Browser Panel

The calculator will open just like below (see Fig. 32).

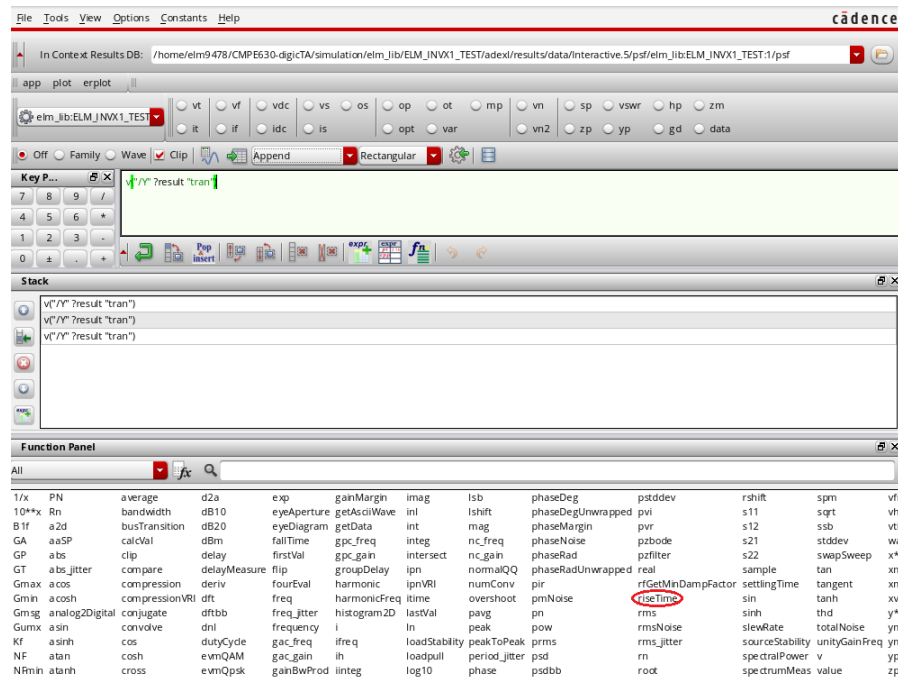


Figure 32: Waveform Window with Hierarchy Browser Panel

Now, let's calculate the pre-layout output signal rise time. From the function panel, select **riseTime**.

Set the initial value of Y to 0, and the Final Value of Y to 1.1 as shown in Fig. 33. For this tutorial, we'll define Rise and Fall time as the time it takes a single pass from 10% to 90% of the signal swing. As such, we'll set the Percent Low to 10 and the Percent High to 90 in the form as shown in Fig. 33.

Note: Percent Low and Percent High are defined differently for **Fall Time calculation**. We will remind you about this later in the lab :)

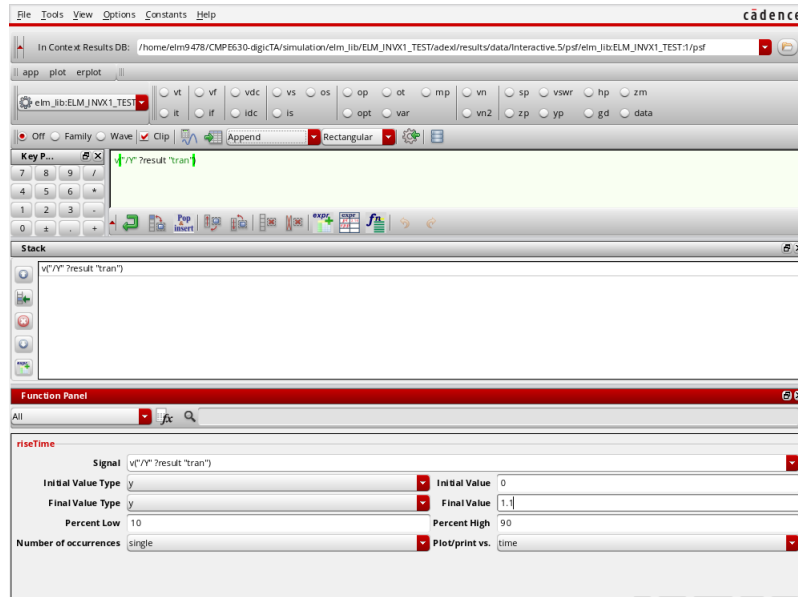


Figure 33: Waveform Calculator Window - Rise Time Calculation Function

Once the form is updated, click Apply to enter the expression into the calculator buffer (see Fig. 34).

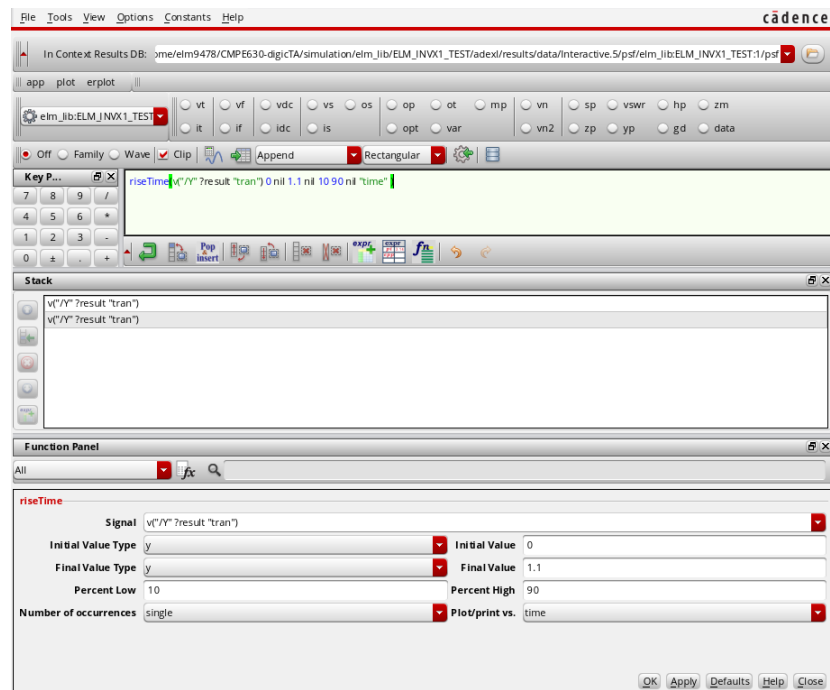


Figure 34: Waveform Calculator Window - Rise Time calculation Function

Once entered, we can evaluate out expression in the calculator buffer by clicking on the evaluate

icon as shown below (see Fig. 35).

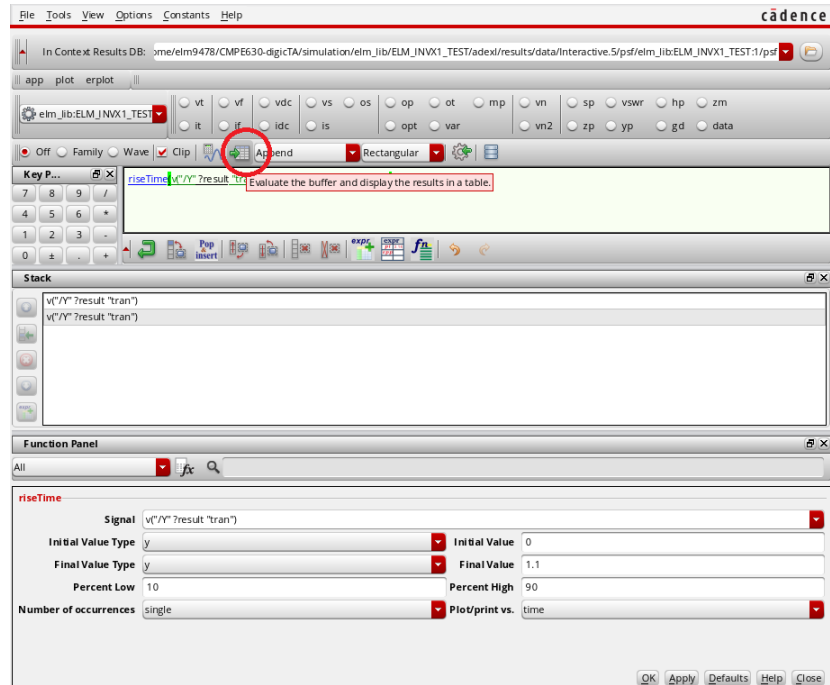


Figure 35: Waveform Calculator Window- Rise Time calculation Function

And the rise time is calculated as shown below (see Fig. 36).

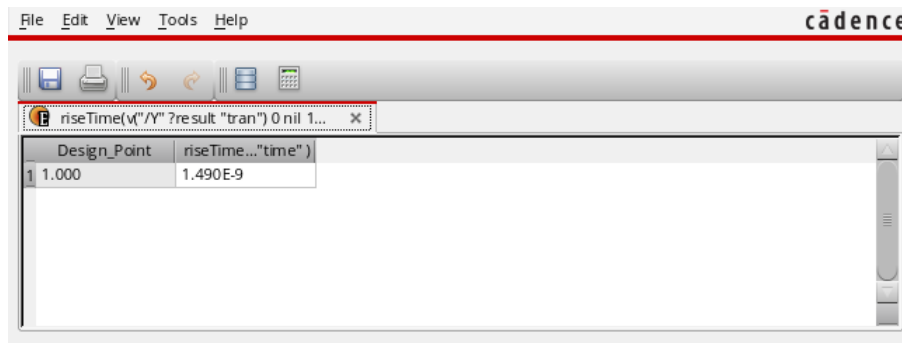


Figure 36: Waveform Calculator Window- Rise Time Calculation Results

Now that we have the rise time value, we can improve our efficiency with the tools by adding this (or any valid calculator expression we desire) to the ADE-XL output setup tab. Return to the calculator and towards the bottom of the window click on the **Expression Editor** tab as shown in Fig. 37.

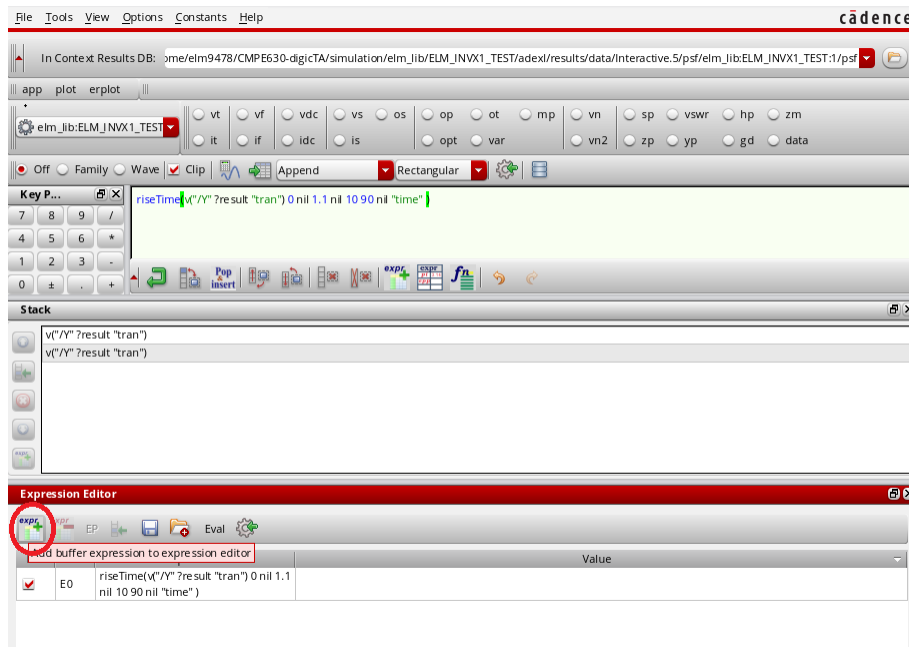


Figure 37: Waveform Calculator Window- Rise Time calculation Expression Editor

The name **E0** is not very descriptive, so let's rename the expression as **RiseTime_Y**

When renaming expressions during future projects, note not all characters are supported, so check the CIW for errors if your expression name is not accepted.

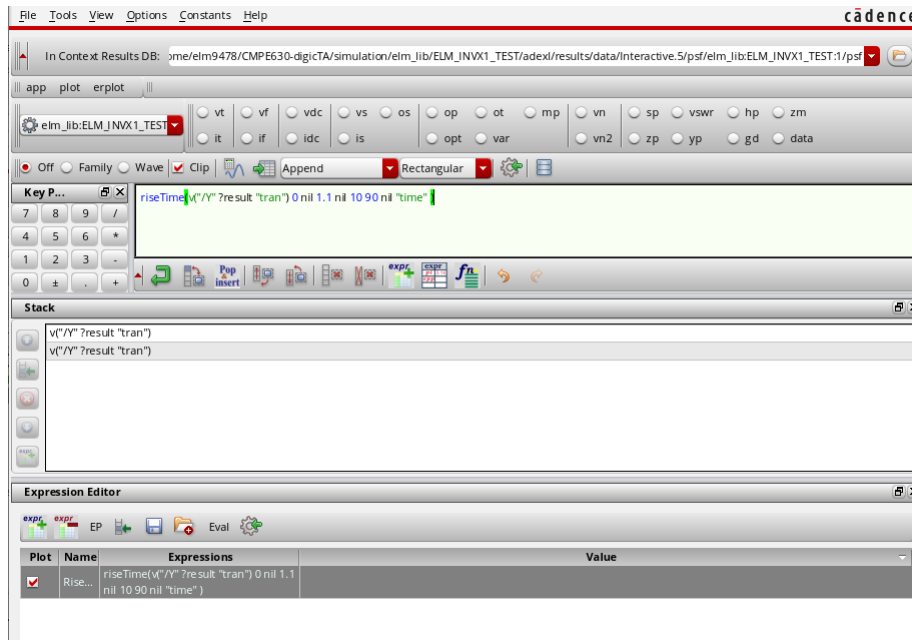


Figure 38: Waveform Calculator Window- Rise Time calculation Expression Editor - Rename Expression

Now Click the little gear icon (see Fig. 39) to **Send selected expressions to ADE outputs.**

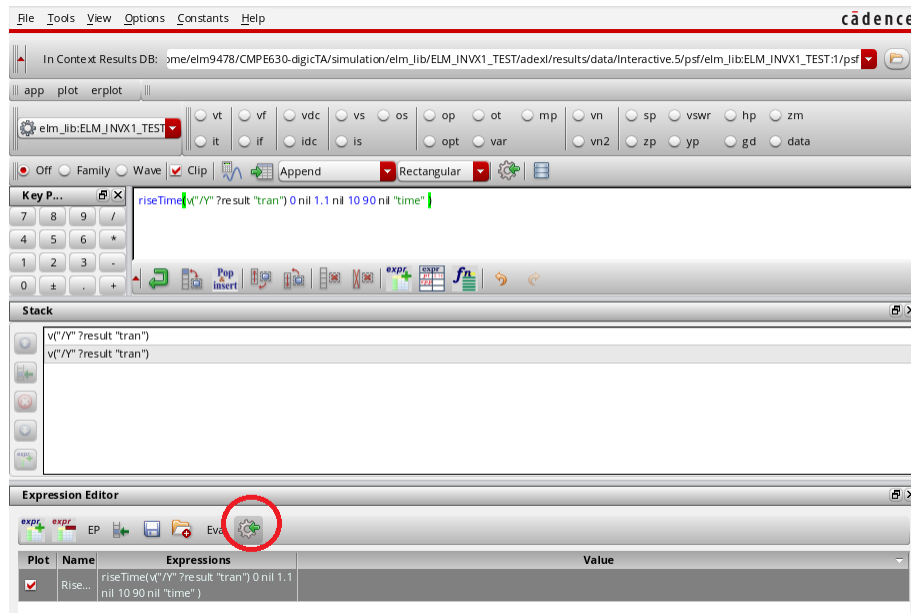


Figure 39: Waveform Calculator Window- Rise Time calculation Expression Editor - Export Expression

As shown in Fig. 40, the rise time calculation expression now appears as an output in the **Outputs Setup** tab; the expression will be evaluated once the simulation runs.

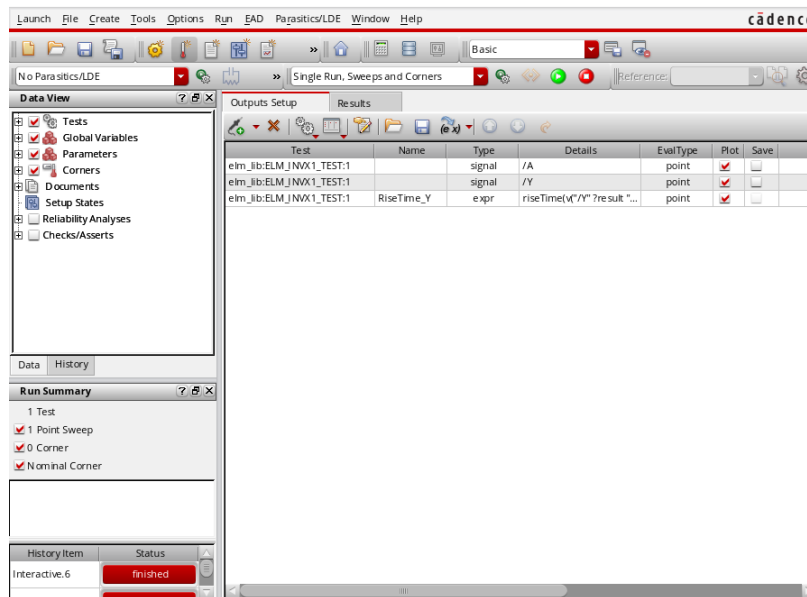


Figure 40: ADE XL Test Editor Window - Outputs Setup with Rise Time Calculation Expression

You are now ready to re-run your transient simulation. Click on the green start icon.

Once the simulation has run, the result of the expression is added to the **Results** tab (see Fig. 41).

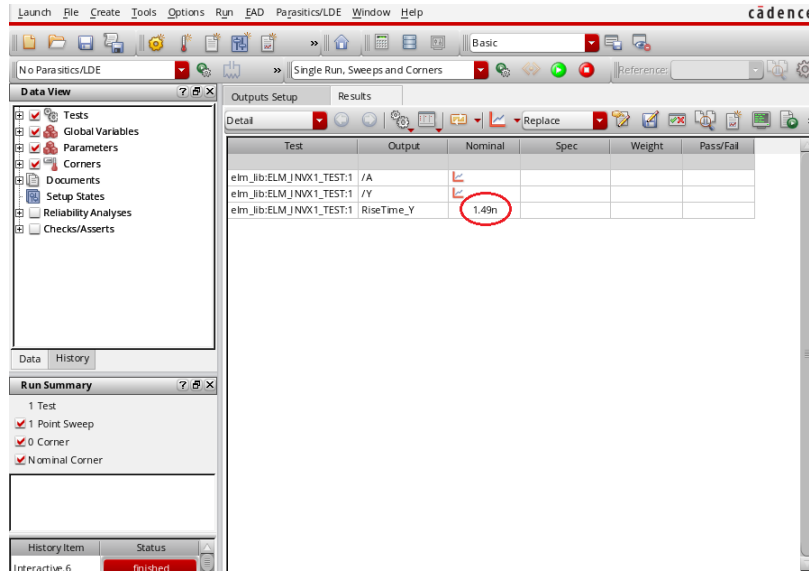


Figure 41: ADE XL Test Editor Window - Results with Rise Time Calculation Result

Then you can close the calculator results, calculator and waveform windows.

Now we want to add DC sweep simulation to plot the Inverter voltage transfer characteristics (VTC) curve. Note that this could be done for most types of gates, but you have to keep all inputs fixed for the one that is swept.

So that we don't have to enter all of the setup again, we'll make a copy of our initial test as shown in Fig. 42:

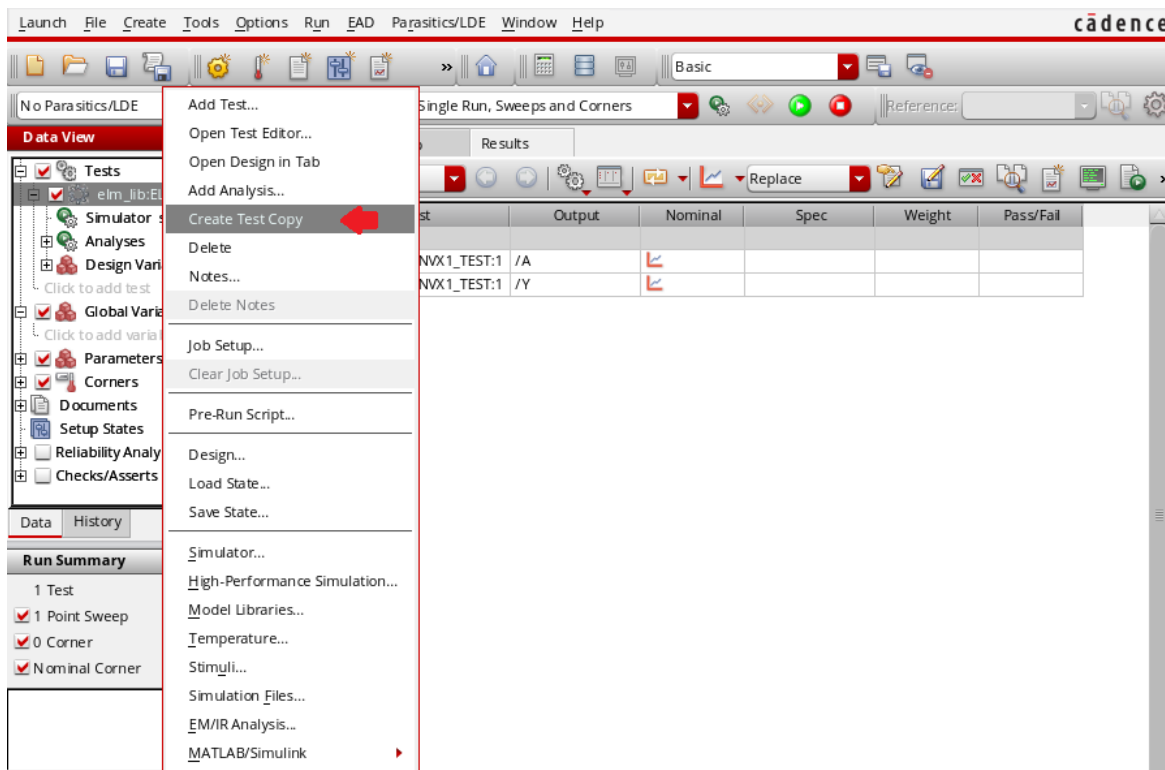


Figure 42: ADE XL Copy Test

Now add a design variable. In your new test copy, expand Design Variables and click to add a variable. The following form (Fig. 43) appears:

The screenshot shows the 'Edit Design Variables Form' dialog box. It has a 'Selected Variable' section on the left and a 'Design Variables' table on the right. The 'Selected Variable' section contains a 'Name' field with the value 'input_A_var' and a 'Value (Expr)' field with the value '0'. Below these fields are buttons for 'Add', 'Delete', 'Change', 'Next', 'Clear', and 'Find'. The 'Design Variables' table has two columns: 'Name' and 'Value'. It contains one row with the name 'input_A_var' and the value '0'. At the bottom of the dialog are buttons for 'OK', 'Cancel', 'Apply', and 'Help'.

Figure 43: Edit Design Variables Form

Add a variable called **input_A_var** within an initial value of 0. Then, click OK.

Now expand **Analysis** and click **Add analysis**. The following form (Fig. 44) appears:

Analysis

☐ tran ☒ dc ☐ ac ☐ noise

☐ xf ☐ sens ☐ dcmatch ☐ acmatch

☐ stb ☐ pz ☐ lf ☐ sp

☐ envlp ☐ pss ☐ pac ☐ pstb

☐ pnoise ☐ pxf ☐ psp ☐ qpss

☐ qpac ☐ qpnoise ☐ qpxf ☐ qpssp

☐ hb ☐ hbac ☐ hbstb ☐ hbnoise

☐ hbsp ☐ hbxf

DC Analysis

Save DC Operating Point ☐

Hysteresis Sweep ☐

Sweep Variable

☐ Temperature

☒ Design Variable Variable Name

☐ Component Parameter

☐ Model Parameter

Sweep Range

☒ Start-Stop Start Stop

☐ Center-Span

Sweep Type

☒ Step Size

☐ Number of Steps

Add Specific Points ☐

Enabled ☒

Figure 44: Configure Analysis form

Select the DC Analysis and add the Sweep Variable as the Design Variable **input_A_var**. Set the Sweep Range Start-Stop to start at 0 and Stop at 1.1. The sweep Type will be Linear with a step Size of 0.01.

Enable this analysis and click OK.

Before we run the simulation though, we need to specify which input we will get the design variable value. Right click on your test and select **Stimuli** (see Fig. 45).

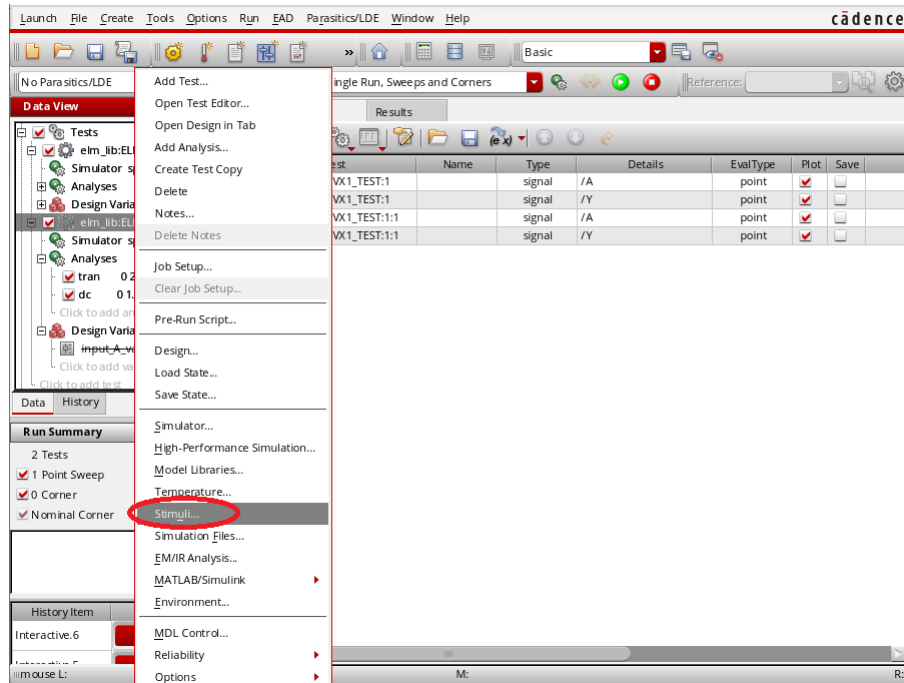


Figure 45: ADE XL - Configure Stimuli

Select input A. Keep all settings and add the Design Variable “**input_A_var**” in the DC voltage box as shown in Fig. 46.

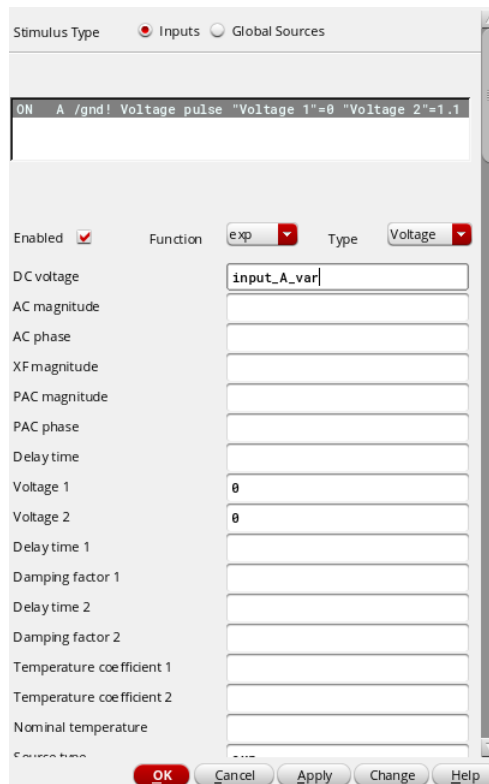


Figure 46: Configure Input Stimuli

This value will be used in the dc analysis. Click OK.

You are ready to run your first DC Sweep simulation. Click on the green start icon.

Once the test is complete, open the waveform viewer and you'll observe both transient and dc sweep analysis waveforms.

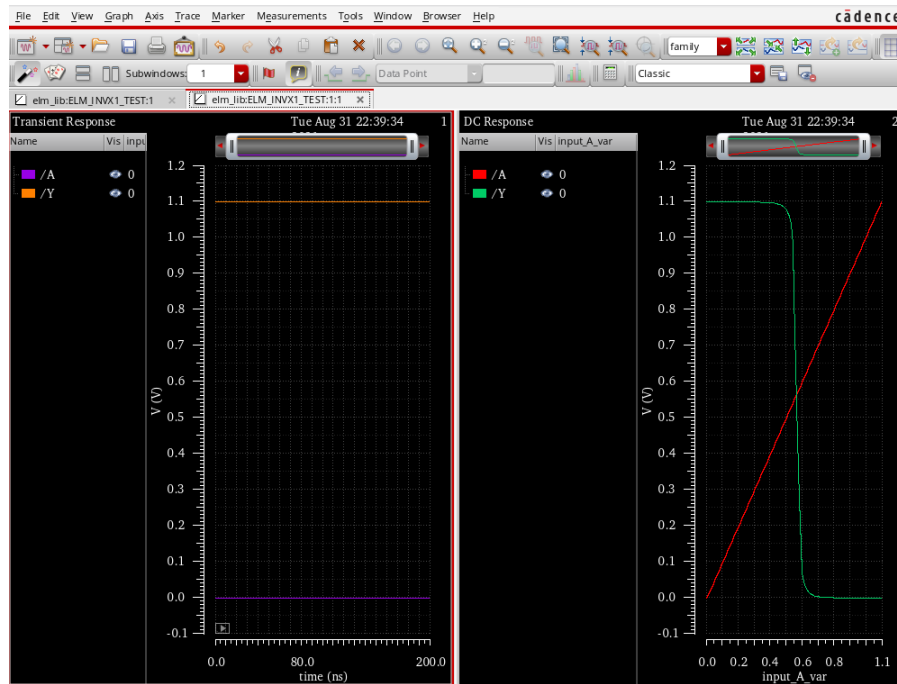


Figure 47: Waveform Results for Transient and DC Analysis

Now, we will measure the midpoint voltage value. Go to **Browser → Results → Open Results**. Select the PSF database. Then select the dc folder just like shown in Fig.48. Again you will see a menu to the left of the screen. Select **Y**, right click and select calculator.

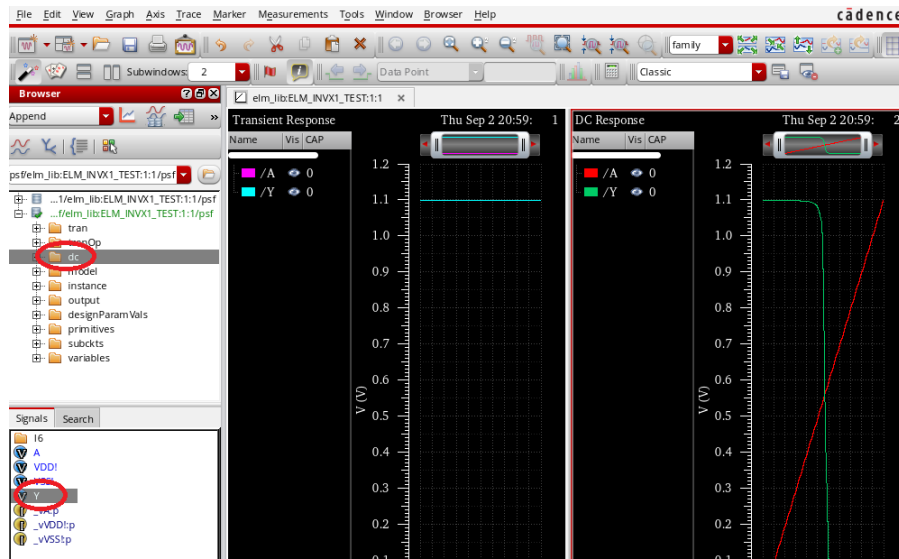


Figure 48: DC Y calculator

In the function panel, look for **Intersect**. Make sure that in the function panel you have signal1 as Y and signal2 as A. Your calculator window should look exactly like in Fig.49

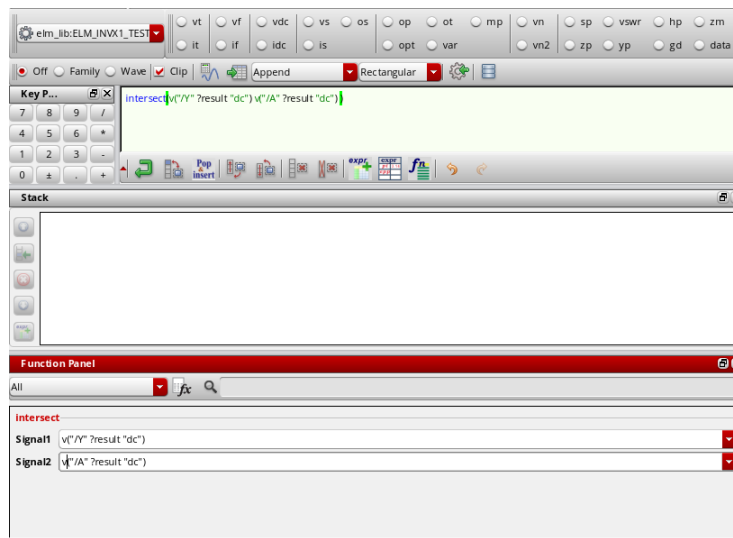


Figure 49: Intersection Function in Function Panel

Now, you may proceed to make this an expression and call it “**midpoint_voltage**”. Now, click the save button. Get A sign-off from the instructor or TA.

2.7 Creating Your Own Tests to Collect Data

We covered a small tutorial on how to create a small test for RiseTime. How you will make tests for more RiseTimes, FallTimes and their delays for nominal and worst cases using a load or no load capacitor (All cases are shown in Table 1. We will be doing the first one together, **you should be looking back at previous steps if you can't remember how to do some things**) and the rest will be on your own. Let's begin!

Table 1: Inverter Results

	Vdd (V)	Temp (C)	Load (fF)	RiseTime	FallTime	T_P,HL	T_P,LH
Nominal	1.1	25	0				
	1.1	25	120				
Worst Case	1	125	0				
	1	125	120				

Create a **copy** of your **very first** test example and rename it **Nominal.RiseTime**. Then, Right click on it and select **Open Test Editor** Just like shown in Fig.50. (We copy because most of the initial set up is already done for you. However you must edit some parameters for each case).

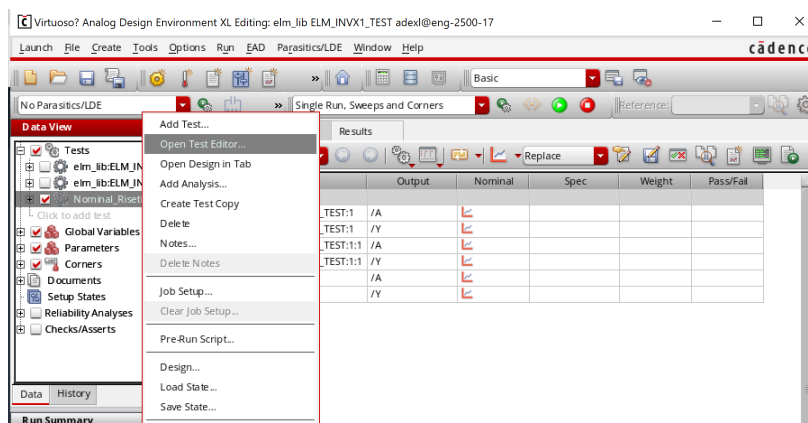


Figure 50: Copy For New Test

Now change the temperature to 25 and the CAP to 0. Then navigate to **Setup → Stimuli** (You don't have to change anything for the nominal tests but when you create the **worst case test** you **will have to change the voltage from 1.1 to 1** for now just close it. *(Reminder: when you make these changes for the worst case, don't forget to always click Apply when you make a change).*

Stimulus Type ☒ Inputs ☐ Global Sources

ON A /gnd! Voltage pulse "Voltage 1"=0 "Voltage 2"=1.1

Enabled ☒ Function pulse Type Voltage

DC voltage

AC magnitude

AC phase

XF magnitude

PAC magnitude

PAC phase

Voltage 1 0

Voltage 2 1.1

Period 50n

Delay time 0

Rise time 50p

Fall time 50p

Pulse width 25n

Temperature coefficient 1

Temperature coefficient 2

Nominal temperature

OK Cancel Apply Change Help

Figure 51: Nominal Stimuli Inputs

Stimulus Type ☐ Inputs ☒ Global Sources

ON VSS! /gnd! Voltage dc "DC voltage"=0
ON VDD! /gnd! Voltage dc "DC voltage"=1.1

Enabled ☒ Function dc Type Voltage

DC voltage 1.1

AC magnitude

AC phase

XF magnitude

PAC magnitude

PAC phase

Temperature coefficient 1

Temperature coefficient 2

Nominal temperature

Source type dc

Noise file name

Number of noise/freq pairs 0

Freq 1

Noise 1

Freq 2

Noise 2

OK Cancel Apply Change Help

Figure 52: Nominal Stimuli Global

You can close out of these and start the simulation using the green button we have been using. Make sure you select the test you want to perform otherwise all tests will run at the same time. After running the simulation, plot it, get the results for **tran**, right click on **Y** and go to the **Calculator**. We will revisit RiseTime again using less steps since it was explained in detail in the example.

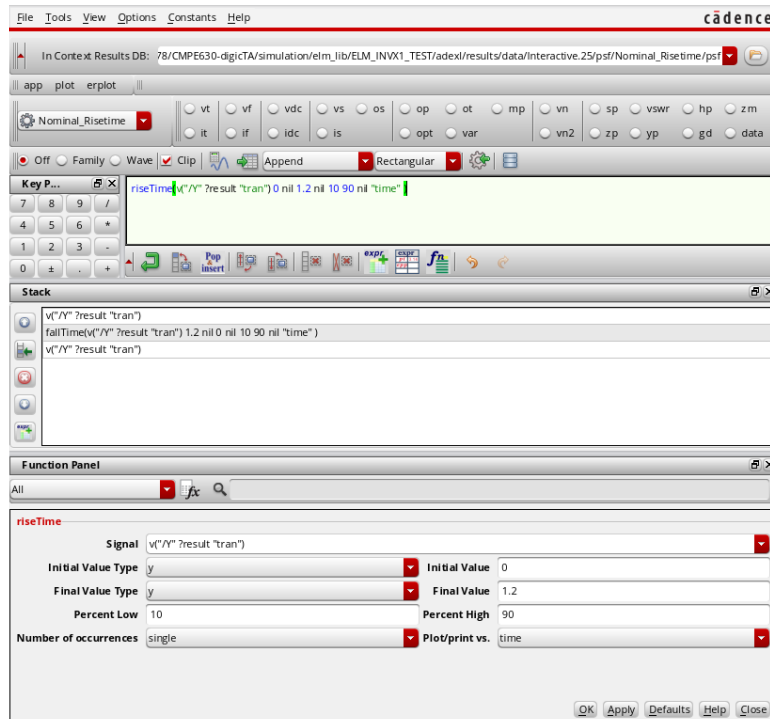


Figure 53: Nominal RiseTime Function

You may check the value using the **Evaluate the buffer and display the results in a table**. Then go to the expressions tab. Deselect any expressions that exist and add the new expression and rename E0 to **Nominal_RiseTime**. If you make a mistake you may erase the expression with the button next to the one that adds.

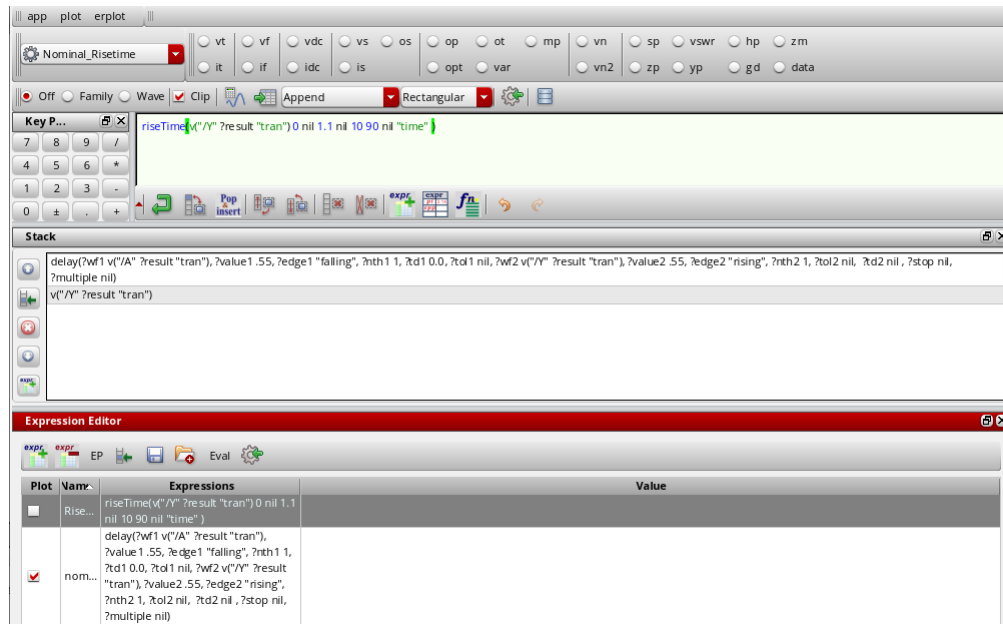


Figure 54: Nominal RiseTime Expression

Follow the same steps for the calculator as mentioned previously (Be sure to not close your calcu-

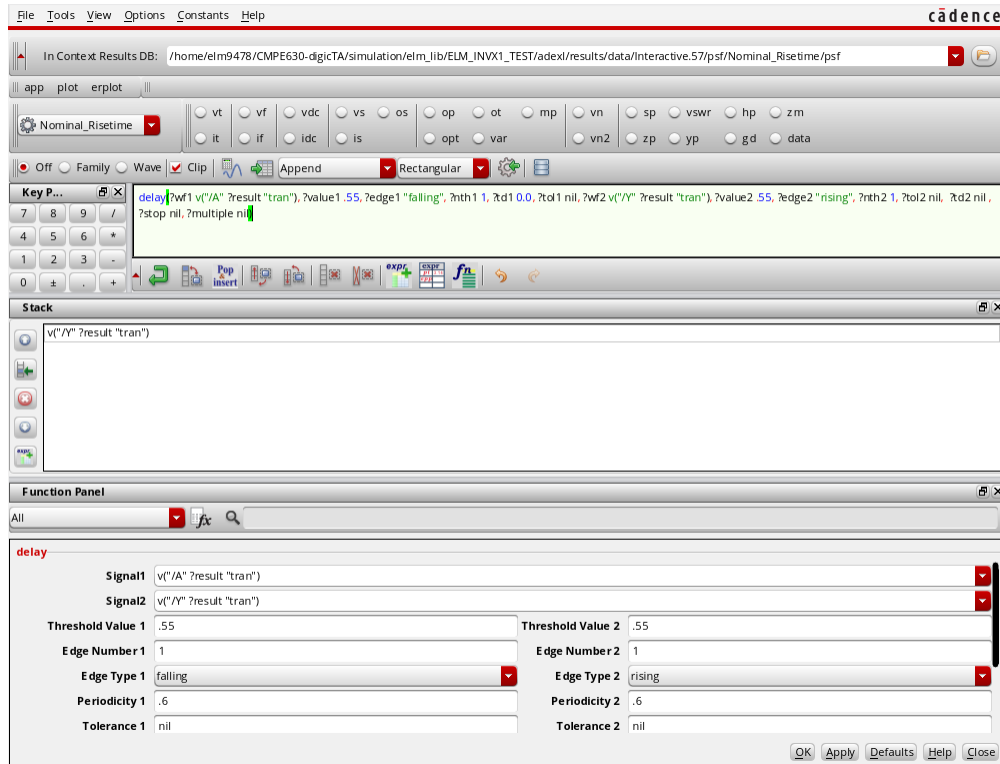


Figure 56: Nominal RiseTime Delay

Follow the same Expression convention. This completed test for the nominal rise time. Your results should mirror the figure below, except that numbers will appear instead of the blue boxes on your screen (see Fig. 57).

Test	Output	Nominal	Spec	Weight	Pass/Fail
Nominal_Risetime	/A				
Nominal_Risetime	/Y				
Nominal_Risetime	nominal_risetime		p		
Nominal_Risetime	nominal_rdelay		p		

Figure 57: Nominal RiseTime Delay Final Test Output

Now, copy this test, rename it to **Nominal_Risetime_load**, and change the **CAP to 120f**. When you run this test, make sure that the CAP variable under Design Variables is checked off, otherwise the simulation will not incorporate it and your results will look identical to the previous step.

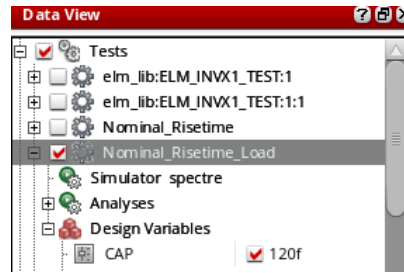


Figure 58: CAP Variable Check

Make sure you have a total of **10 separate tests**. The first one is the example that you use to copy for the next tests you do on your own. The second test is the Midpoint Voltage test and the remaining 8 tests are to fill Table 1. The third test is what we just did. You must fill up Table 1 for sign-off and include it in your report.

- Initial test example (should be completed)
- Midpoint Voltage (should be completed)
- Nominal RiseTime (should be completed)
- Nominal RiseTime with Load
- Nominal FallTime
- Nominal FallTime with Load
- Worst Case RiseTime
- Worst Case RiseTime with Load
- Worst Case FallTime
- Worst Case FallTime with Load

Please check and read reminders on the next page before moving on!

REMINDERS AND ADVICE:

- The load is a CAP with 120 fF. No load is a CAP with 0 fF (i.e., no output capacitor).
 - a. This is not the same as the initial test example in which we used 100 fF!
- The worst case conditions are $V_{dd} = 1\text{ V}$ and $\text{Temp} = 125\text{ C}$.
- FallTime tests measure Fall Time and T_PHL.
 - a. Make sure to use **fallTime** (not riseTime) to measure fall time. Be sure to adjust the delay function for measuring T_PHL (see here).
 - b. **For FallTime, set percent high = 10% and percent low = 90%.**
 - c. For Fall Tests, you need to swap the value of voltage 1 and voltage 2 for the input stimuli A
 - d. If you are curious/having trouble with any function, click the **help** button.
- For this lab, you may want to **disable Global Variable** (It is underneath **Tests** in the dataview window).
 - a. Global variables force the same value on the parameters (i.e., CAP and temp) for all of the tests. We don't want that for this lab since we are testing different load and temperature values.
- Another thing you can do instead of manually creating each test is to make test copies and edit the copies.
 - a. Right click on the test that you want to duplicate and then click on make test copy.
 - b. A new copy of the test should appear below the original one. It will have the same properties as the original one including CAP, temp, inputs stimuli, and all the expression for calculation. For instance, you can make a duplicate of nominal rise time, rename it worst case rise time, and only change the temp and stimuli because everything else should be the same.

The finalized tests should look very similar to Fig. 59 except that you should have your own values. (Please only run the midpoint test and the 8 tests relevant to the table when getting a sign-off).

Take a screenshot of each test with all your results. Your report will require it.

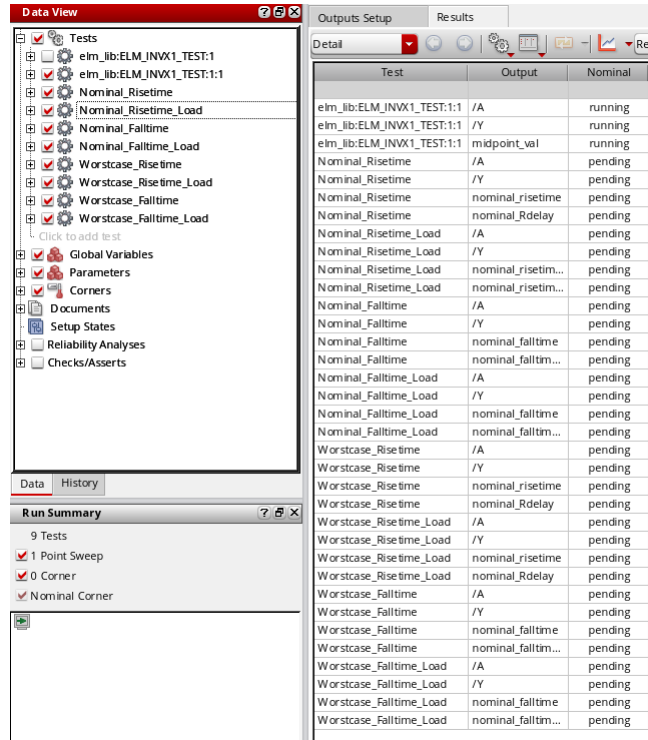


Figure 59: Final Test

Fill up the overall results table (Table 3) using the equations provided in this manual. **The completed Table 2 and 3 are required in the lab report!**

Table 2: Inverter Results

	Vdd (V)	Temp (C)	Load (fF)	RiseTime	FallTime	T _{P,HL}	T _{P,LH}
Nominal	1.1	25	0				
	1.1	25	120				
Worst Case	1	125	0				
	1	125	120				

Table 3: Overall Results

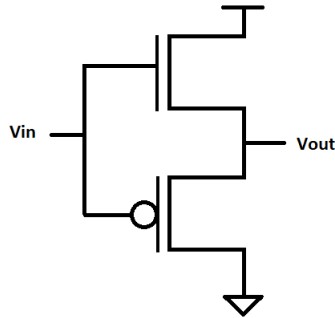
	VDD (V)	Temp (C)	Load (fF)	$F_{input,max}$	$F_{throughput,max}$
Inverter	1	125	0		
	1	125	120		

$$F_{input,max} = \frac{1}{t_{rise} + t_{fall}} \quad (1)$$

$$F_{throughput,max} = \frac{1}{T_{P,HL} + T_{P,LH}} \quad (2)$$

3 Questions

1. What are the regions of operation of an inverter? Describe the mode of operation for each transistor for each region. Use the VTC curve you saw in class to explain.
2. Draw the waveform that would result from a DC sweep of the input of the following circuit.



4 Lab Exercise Report Requirements

These are the requirements for your lab report. Please use the right format for CE lab exercise reports. This is a graduate level course and you should be equipped with a lot of knowledge to write a proper report for this class. Please follow the list of guidelines and apply as much as you can from what you already know on how to explain theory, waveforms, results, behavior, and everything else that should be inside a proper report!

Note: Proper citation is needed for all non-original figures (i.e., those that you have not made yourself).

1. Cover Page

- Name and number of the lab exercise
- Your name, name of the instructor, and name of the TAs
- Submission date

2. Table of contents

3. Abstract

- A description of what you did for this exercise, how you did it, and significant numerical lab results (e.g., input and throughput frequencies)

4. Design Methodology

- Include all necessary figures such as design schematics, symbol designs, and testbench schematics
- Include a discussion of NMOS and PMOS IV characteristics
- Include the inverter voltage transfer characteristics
- Include any relevant knowledge you learned in class about basic transistors and inverters to your report

5. Results and Analysis

- Simulation results include:
 - Transient simulation waveform
 - DC Analysis (sweep) waveform
 - A screenshot of all timing measurements in ADE XL
 - Statement on what midpoint voltage is
- Completed Table 2 and 3 with proper captions (do not forget units!)
- Provide percentage increase or decrease in the maximum operating frequency and the maximum throughput from the loaded to the worst unloaded

6. Conclusion

- Restate important results, how they were obtained, any uncertainties, possible improvements, and any implications.
- This should not be a copy-paste of the Abstract

7. Questions

- Re-write the question(s)
- Answer all of the questions from the “questions” section.

8. References (if needed)

9. Sign-off Sheet

- This is absolutely required! Lab reports will not be graded without a sign-off sheet. Otherwise your grade will be an automatic zero.

5 Signoff Sheet

Students Name:

Section:

Demo	Point Value	Points Earned	Date
Schematic CIW Check	5	5	JE 1/31
Transient Simulation	5	5	↓
DC Sweep Simulation with midpoint calculation	20	20	
Nominal Tests	35		
Worst Case Tests	35		